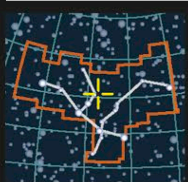
Sb SPIRAL GALAXY

Andromeda Galaxy



ANDROMEDA

CATALOG NUMBERS
M31, NGC 224

DISTANCE
2.5 million light-years

DIAMETER
220,000 light-years

MAGNITUDE 3.4

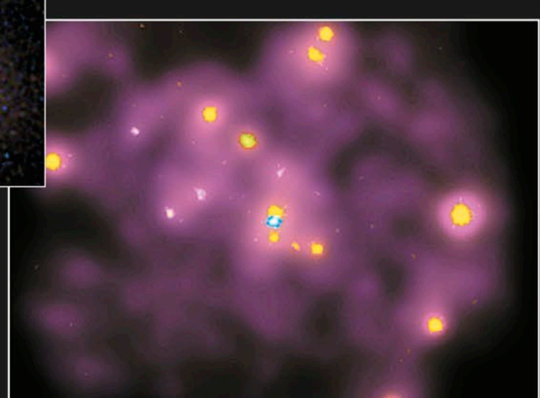
The Andromeda Galaxy (M31) is the closest major galaxy to the Milky Way and the largest member of the Local Group of galaxies. Its disk is twice as wide as our galaxy's.

M31's brightness and size mean it has been studied for longer than any other galaxy. First identified as a "little cloud" by Persian astronomer Al-Sufi (see p.421) in around 964 CE, it was for centuries assumed to be a nebula, at a similar distance to other objects in the sky. Improved telescopes revealed that this "nebula," like many others, had a spiral structure. Some astronomers thought that M31 and other "spiral nebulae" might be solar systems in the process of formation, while others guessed correctly that they were independent systems of many stars. It was in the early 20th century that Edwin Hubble (see p.45) revealed the true nature of M31, at a stroke hugely increasing estimates of the size of the universe (see panel, opposite). Astronomers now know that M31, like the Milky Way, is a huge galaxy attended by a cluster of smaller orbiting galaxies, which occasionally fall inward under M31's gravity and are torn apart.

Despite being intensively studied, the Andromeda Galaxy still holds many mysteries, and it may not be as typical a spiral galaxy as it appears. For example, despite its huge size, it seems to be only half as massive as the Milky Way, with a relatively sparse halo of dark matter. Despite this, M31 contains about a trillion stars, and its supermassive black hole has a mass of 150 million Suns, 40 times the mass of the Milky Way's central black hole. This big difference is surprising, because the black hole is thought generally to reflect the mass of its parent galaxy. Furthermore, studies at different wavelengths have revealed disruption in the galaxy's disk, possibly caused by an encounter with one of its satellite galaxies in the past few million years.

M31 and the Milky Way are moving toward each other, and they should collide and begin to coalesce in around 5 billion years.

DOUBLE CORE
This Hubble close-up shows two distinct concentrations of stars at the very center of Andromeda. The central black hole lies within the fainter area.



CENTRAL BLACK HOLE
This X-ray image of a small area of M31's core shows its central black hole as a blue dot—it is cool and inactive compared to the galaxy's other X-ray sources (yellow dots).